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Paleomagnetic evidence of Tertiary tectonic rotation in west Texas

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ABSTRACT

Paleomagnetic samples obtained from Rawls Formation (26.9–28.3 Ma) lava flows of Bofecillos volcano in west Texas give a paleomagnetic pole at lat 76.5°N, long 348.0°E (α_{95} = 9.4°; N = 20). This pole is displaced significantly from North American reference poles of similar age, implying a clockwise rotation of 21.3° ±8.8°. It is also consistent with other paleomagnetic data from west Texas. We argue that the discordance is a result of vertical-axis crustal rotation in west Texas caused by shear deformation that was probably related to Basin and Range tectonics.

INTRODUCTION

Late Cenozoic tectonic deformation in the western United States has been mostly extensional and has formed the Basin and Range province. Although simple extension should cause the crust to undergo only vertical motion or rotation about a horizontal axis, vertical-axis rotations have been documented, mainly by the use of paleomagnetism to quantify a deflection of the paleo-north direction. Some rotations have resulted from local dip-slip motions in normal-fault zones (Wells and Hillhouse, 1989), but they typically appear to be related to strike-slip movement within the west coast transform fault zone or shear transfer zones in the Basin and Range (e.g., Beck, 1980; Brown and Golombek, 1986; Hudson and Geissman, 1987).

West Texas is at the edge of the Basin and Range province near the boundary of the Rio Grande rift (Fig. 1). East-northeast extension formed north-trending basins in this region and is thought to be the dominant tectonic force (Henry et al., 1991). However, west-northwest strike-slip displacement has occurred on faults within the Texas lineament zone. This ~100-km-wide zone is located along the Rio Grande River, and may have resulted from reactivation of basement structures by the Basin and Range tectonic stress field (Eaton, 1979; Muehlberger et al., 1980). Because fault offsets are small, rotation is thought to be insignificant (e.g., Henry et al., 1991), but this view is not universal (Stevens and Stevens, 1990).

Few paleomagnetic studies have been done in west Texas, and most addressed problems other than tectonics. To document rotation, a paleomagnetic vector is measured for a dated formation and is compared to the direction of the appropriate reference pole of the North American apparent polar wander path (APWP). In 1986 we began a pilot study by sampling the Oligocene Paisano volcano near Alpine at the edge of the Texas lineament zone (Fig. 1). That study gave a paleodeclination about 13° clockwise from the expected direction (Ryan, 1988), but because of data scatter and a small number of independent units sampled, these results alone are not compelling. Thus, in 1990 we sampled Bofecillos volcano in the Big Bend Ranch State Natural Area, in the middle of the Texas lineament zone, to see if our previous results could be substantiated.

BOFECILLOS VOLCANO AND THE TRANS-PECOS VOLCANIC FIELD

The Trans-Pecos volcanic field of west Texas and northern Chihuahua (Fig. 1) is the southeastern extension of Cenozoic Cordilleran volcanism (Henry et al., 1991). Activity in Texas was nearly continuous from 48 to 17 Ma (Henry and McDowell, 1986). Until about 31 Ma the volcanism was alkalic and alkalic-calcic, commonly explosive, and probably generated by the subduction of the Farallon plate under western North America (Coney and Reynolds, 1977). Later volcanism was volumetrically minor, basaltic in composition, and apparently a response

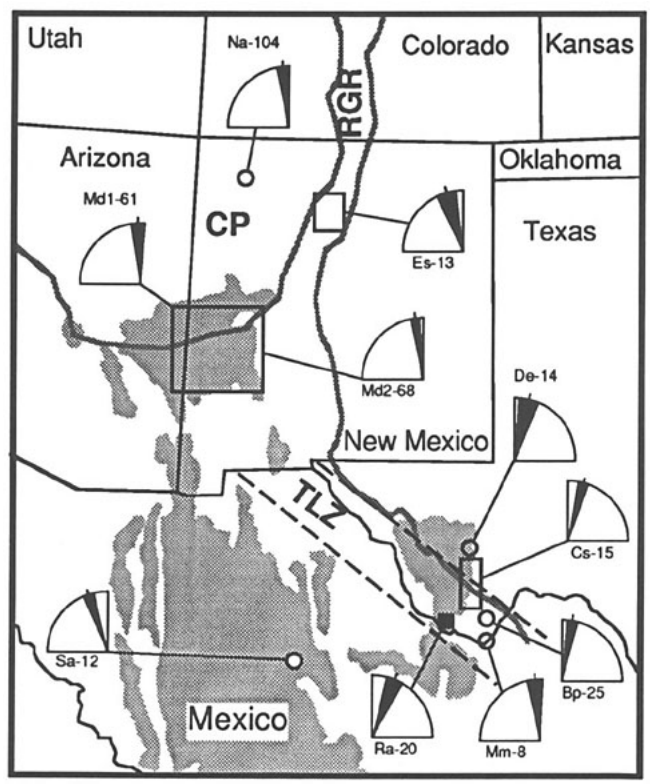


Figure 1. Location map of study area with paleomagnetic results from early Tertiary rocks in west Texas and environs. Solid black square shows location of Bofecillos Mountains. Open circles and boxes show locations of other study areas. Quadrants with black wedges represent paleomagnetic discordance at each study site (observed minus expected paleodeclination). Tick on quadrant perimeter denotes amount of discordance; black wedge shows 95% confidence limits. Site labels denote sampled formation and number of samples (Bp = Black Peaks; Cs = Cottonwood Springs; De = Decie, Es = Espinaso; Md1, Md2 = Mogollon-Datil; Mm = Mariscal Mountain; Na = Nacimiento; Ra = Rawls; Sa = San Antonio). Gray stippled patches indicate volcanic fields. Thick gray line shows approximate limits of Basin and Range province. CP = Colorado Plateau; RGR = Rio Grande rift; TLZ = Texas lineament zone.

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to Basin and Range extension (Henry et al., 1991). The Bofecillos stratovolcano formed during the transition and erupted basaltic to intermediate composition Rawls Formation lavas over a wide area of the southern Big Bend (Fig. 2). Six K-Ar dates for the Rawls Formation range from 26.9 to 28.3 Ma (Henry and McDowell, 1986).

METHODS

Paleomagnetic samples were collected at 25 Rawls Formation sites (Fig. 2); the sampling was guided by geologic maps of the region (Erickson, 1953; McKnight, 1970). The sites included 19 in latite porphyry flows, four in basalt flows, one in a rhyolite flow, and another in a fine-grained andesite flow. From seven to 19 minicores, 2.5 cm in diameter, were obtained at each site (Table 1) with a portable rock drill; the minicores were oriented by sun compass. To avoid lightning-strike remagnetization, low-relief outcrops and dry stream beds were preferred sampling sites.

Stepwise thermal or alternating-field (AF) demagnetization was performed on each sample to remove unstable overprint magnetizations and to isolate the characteristic remanent magnetization (ChRM), the direction of which is assumed to represent the magnetic field at the time the rock was cooled after eruption. To determine which demagnetization method to use, several samples from each site were AF demagnetized and then stepwise remagnetized with an isothermal remanent magnetization (IRM) using a pulse magnetizer that produced fields of up to 1200 mT. IRM acquisition curve shapes were used to determine whether high-coercivity magnetic minerals were present. Samples with high-coercivity behavior were treated with thermal demagnetization, typically with 12 steps up to 700 °C. Other samples were treated with AF demagnetization, usually with 10 steps up to 100 mT.

An orthogonal-vector plot was made for each sample to look for univectorial decay toward the origin, which would indicate that the ChRM had been isolated. ChRM directions were calculated by averaging three consecutive demagnetization steps from the univectorial decay sequence. To obtain a site mean direction, Fisher statistics were used to average sample directions. Virtual geomagnetic poles (VGPs) were calculated from each site mean and averaged to obtain a paleomagnetic pole, again using Fisher statistics, but omitting as unreliable site means with α_{95} values greater than 26°. Calculations of expected paleomagnetic directions, discordances with expected directions, and the errors of both were calculated using the methods described by Beck (1980) and Demarest (1983).

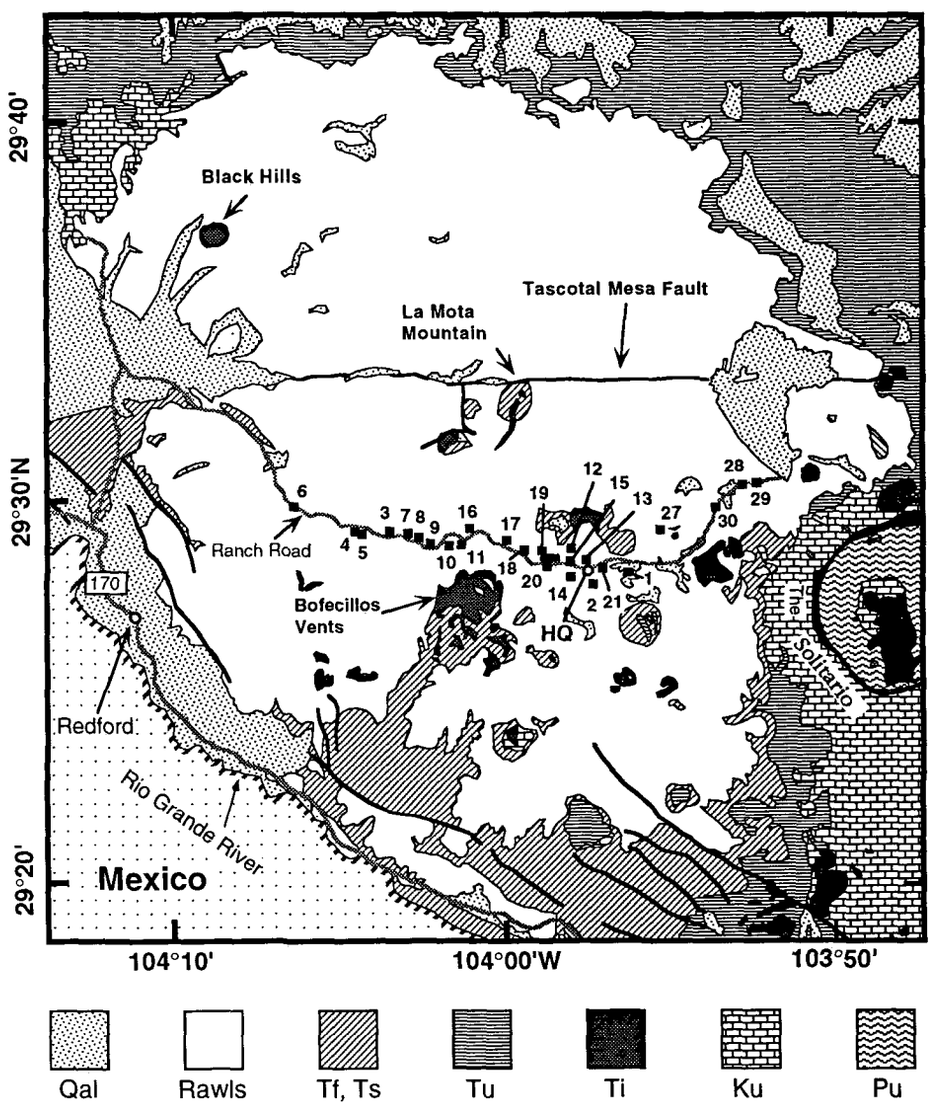


Figure 2. Simplified geologic map showing Rawls Formation and sampling site locations. Numbers identify sites (see Table 1). Heavy black lines are faults. Thick gray lines are roads. Qal = Quaternary alluvium; Rawls = Rawls Formation; Tf and Ts = Fresno and Santana Tuff Formations (Bofecillos Volcano products); Tu = Tertiary volcanic rocks; Ti = Tertiary intrusive rocks; Ku = Cretaceous limestones; Pu = Paleozoic rocks. Map redrawn from Barnes (1979).

RESULTS

Rawls Formation samples were generally strongly magnetic, with natural remanent magnetization (NRM) ranging from 4.0×10^{-3} to 5.2×10^{2} A/m; the logarithmic mean was 7.3×10^{-1} A/m. Three (RF16, RF17, and RF20) of the four most strongly magnetic outcrops showed a large scatter in ChRM directions, probably because of lightning remagnetization. IRM acquisition tests from 17 sites displayed saturation in low applied fields (<0.3 T), implying that magnetite is the dominant magnetic mineral. Of the remaining sites, three showed high coercivities, indicating a magnetic carrier such as hematite. During thermal demagnetization most such samples retained a significant magnetization at 650 °C, consistent with the 680 °C Néel temperature of hematite. Six sites had intermediate behaviors, suggesting a mix of low-and high-coercivity minerals.

Overprint magnetizations were typically small and were removed with 10–20 mT of AF or 150–250 °C of thermal demagnetization (Fig. 3). Of 25 sites, 21 gave reliable results indicated by mean directions with α_{95} values of 26° or less (Table 1); 12 had α_{95} values less than 10°. Nearly all sites had a reversed polarity magnetization. Only one (RF19) was clearly of normal polarity, whereas three (RF08, RF16, and RF18) gave reversed declinations and normal inclinations, perhaps recording transitional field directions.

An average Northern Hemisphere paleomagnetic pole was calculated at 76.5°N, 348.0°E with an α_{95} of 9.4°. As seen in Figure 4, this pole is significantly different from the 30 Ma North America APWP pole of Harrison and Lindh (1982) (84.7°N, 157.7°E, α_{95} = 3.0), as well as the more recent 20–40 Ma pole of Diehl et al. (1988) (81.5°N, 147.3°E, α_{95} = 2.4°). Both reference poles imply that Tertiary rocks from west Texas should record westward paleodeclinations, 354.0° ±2.7° and 351.0° ±2.2° for

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the 30 and 20–40 Ma poles, respectively, rather than the observed Rawls mean paleodeclination of 15.3° ±8.4°. Comparing the Rawls and reference poles, clockwise discordances of 21.3° ± 8.8° and 24.3 ±8.6° are found relative to the 30 and 20–40 Ma poles. Because the 30 Ma pole is better documented and gives the lesser, more conservative discordance, we prefer the 21.3° value for subsequent discussion.

DISCUSSION AND CONCLUSIONS

The discordance between the observed and expected paleodeclinations implies that the crust in the vicinity of Bofecillos volcano has rotated 21.3° clockwise since Oligocene time. Moreover, the observed paleoinclination, 46.9° ±7.4°, is indistinguishable from the expected inclination derived for the study site from the reference pole, 47.5° ±3.5°, indicating a vertical-axis rotation. Before we draw further inferences, it is worth considering whether the discordance could result from systematic error.

A spurious Rawls pole or reference pole could cause the observed declination difference. Possible biasing factors for the Rawls results are (1) incompletely removed overprint magnetization, (2) remagnetization at a later time, (3) incompletely averaged secular variation, or (4) tectonic tilting. For this study, all these can be discounted.

An overprint is unlikely because all samples received a detailed demagnetization treatment and most displayed simple behavior with univectorial decay. A remagnetization is unlikely because there has been no subsequent reheating in the study area (McKnight, 1970), nor is there evidence that weathering has imposed a new magnetization. For example, samples containing hematite (a possible authigenic mineral) gave directions indistinguishable from those of samples dominated by magnetite. Moreover, a recent remagnetization would be unlikely to

TABLE 1. PALEOMAGNETIC DATA FOR RAWLS FORMATION SITES

Site	Location lat (°N)	Location long (°W)	n/N	Demag	ChRM direction Inc (°)	ChRM direction Dec (°)	α_{95}	k	Pol	VGP lat (°N)	VGP lon (°W)
01	29.468	103.944	6/8	AF	-62.9	203.0	16.1	18.3	R	66.3*	301.0
02	29.467	103.959	7/8	AF	-49.1	206.9	9.7	39.4	R	66.7*	338.1
03	29.483	104.108	4/8	TH	-32.2	165.5	10.0	86.0	R	72.1*	126.9
04	29.489	104.073	7/8	AF	-36.6	172.1	3.2	360.5	R	78.4*	116.1
05	29.488	104.072	6/8	AF	-31.2	176.9	7.2	88.6	R	77.0*	89.3
06	29.501	104.059	7/8	AF	-59.6	221.3	5.3	128.6	R	54.8*	316.6
07	29.486	104.053	5/7	AF	-5.0	177.4	12.1	40.8	R	62.9*	81.6
08	29.483	104.046	8/9	AF	15.8	187.3	21.6	7.5	?	51.8	64.3
09	29.481	104.038	5/9	TH	-57.8	211.5	15.3	26.0	R	62.6*	318.4
10	29.483	104.032	7/8	AF	-54.4	170.3	10.1	36.5	R	80.1*	201.8
11	29.483	104.027	7/12	TH	-54.5	183.1	18.0	12.2	R	83.9*	280.4
12	29.478	103.974	9/19	TH	-48.7	175.4	2.8	333.7	R	86.0*	168.9
13	29.471	103.958	6/18	TH	-50.1	177.8	3.0	489.3	R	87.7*	202.8
14	29.471	103.963	9/10	AF	34.8	74.5	82.1	1.4	?	22.4	335.9
15	29.472	103.964	7/9	TH	-45.5	177.5	8.0	58.0	R	86.7*	117.7
16	29.484	104.018	5/12	AF	34.3	200.3	30.5	7.3	?	37.9	51.4
17	29.483	104.005	8/10	TH	31.2	242.4	89.9	1.1	?	14.1	15.0
18	29.476	103.988	5/8	TH	16.1	221.8	30.2	7.4	?	34.9	22.5
19	29.474	103.982	8/12	AF	32.4	11.4	3.9	199.7	N	74.2*	32.3
20	29.473	103.978	3/7	AF	-64.3	222.1	10.2	40.8	R	53.4*	307.3
21	29.475	103.949	7/8	TH	-42.2	193.4	6.0	102.9	R	77.0*	6.0
27	29.485	103.928	6/12	AF	-18.3	226.7	3.2	443.4	R	42.0*	1.0
28	29.509	103.881	3/8	AF	-31.4	230.3	19.7	40.3	R	42.5*	349.8
29	29.518	103.874	8/9	AF	-39.9	219.1	4.1	182.1	R	54.4*	347.8
30	29.497	103.893	7/7	AF	-43.1	217.5	25.7	6.5	R	56.5*	344.5
Mean†	29.5	104.0	20/25		46.9	15.3	9.4	13.2		76.5	348.0

Note: *N* = number of samples; *n* = number of samples for mean direction calculation; *Demag* = demagnetization treatment, *AF* = alternating field ,*TH* = thermal; *ChRM* = characteristic remanent magnetization; *Inc* = mean inclination; *Dec* = mean declination; α_{95} = radius of 95% confidence circle of mean; *k* = precision parameter; *Pol* = polarity, *N* = normal, *R* = reversed; *VGP* = virtual geomagnetic pole.

*Indicates VGP used in mean pole calculation; anti-poles used for R directions.

†Mean: numbers are average for Rawls Formation study. *n/N* = number of site means used in pole calculation/number of total sites; *Inc, Dec* = mean inclination and declination determined from pole location; α_{95}, *k* = 95% confidence circle radius and precision parameter for pole.

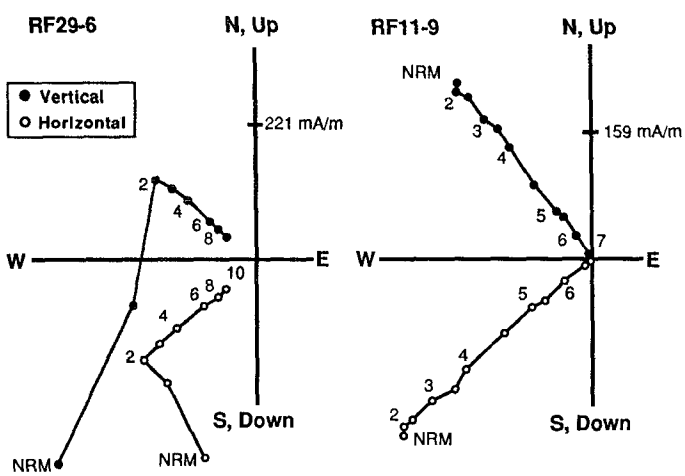


Figure 3. Orthogonal vector plots showing behavior of typical Rawls samples during alternating field (left) and thermal (right) demagnetization. Numbers on left plot are field strength in tens of milliTeslas; numbers on right plot are hundreds of degrees Centigrade.

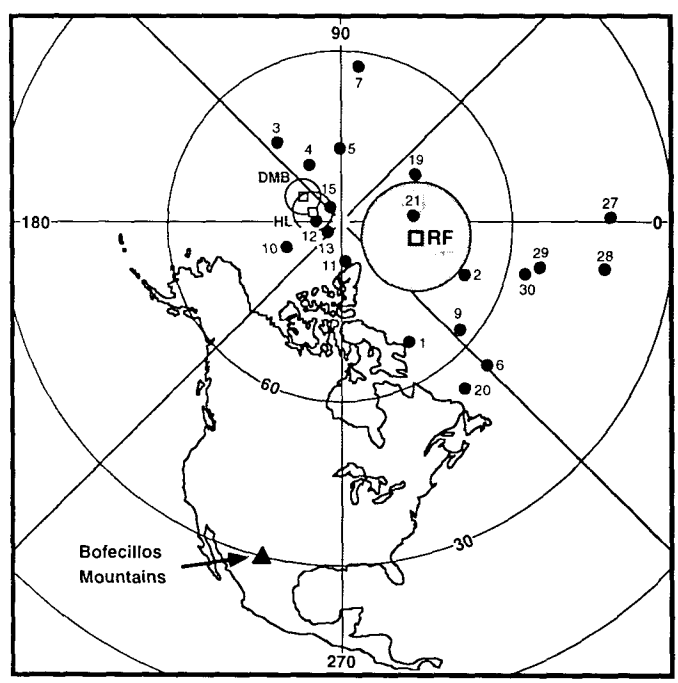


Figure 4. Comparison of Rawls Formation paleomagnetic poles with North American reference poles and other early to mid-Tertiary paleomagnetic data from study region. Solid circles are virtual geomagnetic poles for Rawls sites, identified by site number. Large open square and surrounding circle are Rawls Formation (RF) mean pole and 95% confidence circle. Small open squares and surrounding circles are 30 Ma (Harrison and Lindh, 1982), and 20–40 Ma (Diehl et al., 1988) North American reference poles, denoted by HL and DMB, respectively.

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change the declination from west to east, as is observed. Incomplete secular variation averaging is probably not the problem, because although a comparison of site mean directions showed some serial correlation of sites near one another, the between-site scatter implies that there are at least 15 independent site estimates of the paleofield direction. Tilting does not seem to be the cause because the measured paleoinclination has the expected value. Although an APWP inaccuracy is possible because some Tertiary data are from areas known to be affected by Basin and Range tectonics, the APWP trend is consistent with the drift of North America relative to the hotspots (Morgan, 1983).

We conclude that the Rawls pole indicates a 21.3° clockwise rotation of west Texas crust relative to the North American craton. This is unexpected because it is thought that extension has been the primary mode of Tertiary deformation in this region (Barker, 1977; Henry et al., 1991). However, evidence has been cited for strike-slip motion in west Texas (e.g., Eaton, 1979; Muehlberger et al., 1980; Stevens and Stevens, 1990); the largest west Texas earthquake, the 1931 Valentine temblor, gave a focal mechanism solution that is almost entirely strike slip (Dumas et al., 1980). Similar paleomagnetic results from other areas in western North America have been interpreted as tectonic rotation resulting from a shear couple distributed across a faulted zone (Brown and Golombek, 1986; Ron et al., 1984; Hudson and Geissman, 1987). Thus, our results suggest that a right-lateral shear zone exists in west Texas.

With available data, we cannot delimit adequately the timing and areal extent of the rotation. It occurred sometime after about 27–28 Ma, when the Rawls Formation was deposited. Other paleomagnetic data suggest that the clockwise rotation was relatively widespread in west Texas (Fig. 1). Data from the Oligocene Paisano volcano (Ryan, 1988) imply a 12.7° ±11.5° discordance; igneous sites from Alpine to Big Bend National Park give a similar result, 14.5° ±6.5° (Winters, 1967); and sediments from the Paleocene Black Peaks Formation indicate a 9.2° ±6.1° discordance (Rapp et al., 1983). However, data from Mariscal Mountain, located within the Texas lineament zone at the southern tip of the Big Bend, show no significant discordance, −4.1° ±6.8° (Harlan et al., 1987). Although this last result is not compelling because of the small number of units sampled (N = 8), it suggests that tectonic complexities exist within the shear zone.

Paleomagnetic data from surrounding regions give a different picture, and imply that the clockwise rotation was probably limited to west Texas. Discordances from the eastern edge of the Colorado Plateau in western New Mexico (Nacimiento Formation, −4.0° ±6.8°, Butler and Taylor, 1978; Mogollon-Datil Volcanics, −2.6° ±5.1°, Diehl et al., 1988; −7.1° ±4.8°, McIntosh, 1991) are consistent with no rotation (Fig. 1). However, discordances from the northern Rio Grande rift in New Mexico (−15.8° ±10.4°, Brown and Golombek, 1986) and the Sierra Madre Oriental in Chihuahua, Mexico (−18.5° ±7.2°, Urrutia-Fucugauchi, 1981) imply counterclockwise rotation (Fig. 1). It is clear that the region around west Texas is tectonically complex. We postulate that the clockwise rotation is a Basin and Range phenomenon limited to the Texas lineament zone in a manner similar to that proposed by Eaton (1979), Muehlberger et al. (1980), and Stevens and Stevens (1990). A key question to be answered by future research concerns the size and continuity of rotational domains in this zone.

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REFERENCES CITED

Barker, D.S., 1977, Northern Trans-Pecos magmatic province: Introduction and comparison with the Kenya rift: Geological Society of America Bulletin, v. 88, p. 1421–1427.

Barnes, V.E., 1979, Geologic atlas of Texas, Emory Peak–Presidio sheet: Austin, University of Texas, Bureau of Economic Geology, scale 1:250,000.

Beck, M.E., Jr., 1980, Paleomagnetic record of plate-margin tectonic processes along the western edge of North America: Journal of Geophysical Research, v. 85, p. 7115–7131.

Brown, L.L., and Golombek, M.P., 1986, Block rotations in the Rio Grande rift, New Mexico: Tectonics, v. 5, p. 423–438.

Butler, R.F., and Taylor, L.H., 1978, A middle Paleocene paleomagnetic pole from the Nacimiento Formation, San Juan basin, New Mexico: Geology, v. 6, p. 495–498.

Coney, P.J., and Reynolds, S.J., 1977, Cordilleran Benioff zones: Nature, v. 270, p. 403–406.

Demarest, H.H., Jr., 1983, Error analysis for the determination of tectonic rotation from paleomagnetic data: Journal of Geophysical Research, v. 88, p. 4321–4328.

Diehl, J.F., McClannahan, K.M., and Bornhorst, T.J., 1988, Paleomagnetic results from the Mogollon-Datil volcanic field, southwestern New Mexico, and a refined mid-Tertiary reference pole for North America: Journal of Geophysical Research, v. 93, p. 4869–4879.

Dumas, D.B., Dorman, H.J., and Latham, G.V., 1980, A reevaluation of the August 13, 1931 Valentine, Texas earthquake: Seismological Society of America Bulletin, v. 70, p. 1171–1180.

Eaton, G.P., 1979, A plate tectonic model for late Cenozoic crustal spreading in the western United States, *in* Reicker, R.E., ed., Rio Grande rift: Tectonics and magmatism: Washington, D.C., American Geophysical Union, p. 7–32.

Erickson, R.L., 1953, Stratigraphy and petrology of the Tascotal Mesa quadrangle, Texas: Geological Society of America Bulletin, v. 64, p. 1353–1386.

Harlan, S.S., Geissman, J.W., Price, J.G., Henry, C.D., Onstott, T.C., and Oldow, J., 1987, Paleomagnetic documentation of the cessation of Chihuahuan fold thrust belt deformation, SW Texas: Geological Society of America Abstracts with Programs, v. 19, p. 693–694.

Harrison, C.G.A., and Lindh, T., 1982, A polar wandering path for North America during the Mesozoic and Cenozoic: Journal of Geophysical Research, v. 87, p. 1903–1920.

Henry, C.D., and McDowell, F.W., 1986, Geochronology of magmatism in the Tertiary volcanic field, Trans-Pecos Texas, *in* Price, J.G., et al., eds., Igneous geology of Trans-Pecos Texas, Guidebook 23: Austin, University of Texas, Bureau of Economic Geology, p. 99–122.

Henry, C.D., Price, J.G., and James, E.W., 1991, Mid-Cenozoic stress evolution and magmatism in the southern Cordillera, Texas and Mexico: Transition from continental arc to intraplate extension: Journal of Geophysical Research, v. 96, p. 13,545–13,560.

Hudson, M.R., and Geissman, J.W., 1987, Paleomagnetic and structural evidence for middle Tertiary counterclockwise rotation in the Dixie Valley region, west-central Nevada: Geology, v. 15, p. 638–642.

McIntosh, W.C., 1991, Evaluation of paleomagnetism as a correlation criterion for Mogollon-Datil ignimbrites, southwestern New Mexico: Journal of Geophysical Research, v. 96, p. 13,459–13,483.

McKnight, J.F., 1970, Geology of Bofecillos Mountains area, Trans-Pecos Texas: Austin, University of Texas, Bureau of Economic Geology, Geologic Quadrangle Map 37, scale 1:48,000.

Morgan, W.J., 1983, Hotspot tracks and the early rifting of the Atlantic: Tectonophysics, v. 94, p. 123–139.

Muehlberger, W.R., Belcher, R.C., and Goetz, L.K., 1980, Quaternary faulting in trans-Pecos Texas: Geology, v. 6, p. 337–340.

Rapp, S.D., McFadden, B.J., and Shiebout, J.A., 1983, Magnetic polarity of the early Tertiary Black Peaks Formation, Big Bend National Park, Texas: Journal of Geology, v. 91, p. 555–572.

Ron, H., Fruend, R., Garfunkel, Z., and Nur, A., 1984, Block rotation by strike-slip faulting: Structural and paleomagnetic evidence: Journal of Geophysical Research, v. 89, p. 6256–6270.

Ryan, D., 1988, Paleomagnetism of Paisano volcano, Texas [M.S. thesis]: College Station, Texas A&M University, 153 p.

Stevens, J.B., and Stevens, M.S., 1990, Allostratigraphy and major structural-tectonic events along and near the Rio Grande, Trans-Pecos Texas, and adjacent Chihuahua and Coahuila, Mexico, *in* Dickerson, P.W., et al., eds., Geology of the Big Bend and Trans-Pecos Texas (South Texas Geological Society 74th Annual Convention guidebook): Tulsa, Oklahoma, American Association of Petroleum Geologists, p. 73–116.

Urrutia-Fucugauchi, J., 1981, Paleomagnetic evidence for tectonic rotation of northern Mexico and the continuity of the Cordilleran orogenic belt between Nevada and Chihuahua: Geology, v. 9, p. 178–183.

Wells, R.E., and Hillhouse, J.W., 1989, Paleomagnetism and tectonic rotation of the lower Miocene Peach Springs Tuff: Colorado Plateau, Arizona, to Barstow, California: Geological Society of America Bulletin, v. 101, p. 846–863.

Winters, M.D., 1967, A paleomagnetic study of early Tertiary basalts in west Texas [M.S. thesis]: Houston, Texas, Rice University, 80 p.

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